

Baoyu Zhou

CONTACT INFORMATION	BYENG 324 699 S Mill Ave Tempe, AZ 85281	+1 (551) 228-9835 baoyu.zhou@asu.edu https://baoyuzhou18.github.io
RESEARCH INTERESTS	Large-scale Continuous Optimization, Constrained Optimization under Uncertainty, Quantum Computing and Optimization, Optimization Methods for Machine Learning	
ACADEMIC POSITIONS	Assistant Professor Arizona State University , Tempe, AZ <i>School of Computing and Augmented Intelligence</i>	July 2024 – present
	Research Affiliate Mayo Clinic , Phoenix, AZ <i>Department of Radiation Oncology</i>	June 2026 – present
	Postdoctoral Research Fellow University of Michigan , Ann Arbor, MI <i>Department of Industrial and Operations Engineering</i>	Aug. 2023 – Apr. 2024
	Principal Researcher University of Chicago , Chicago, IL <i>Booth School of Business</i>	Aug. 2022 – July 2023
EDUCATION	Lehigh University Ph.D. in Industrial & Systems Engineering Advisor: Frank E. Curtis Doctoral Committee: Albert S. Berahas, Jorge Nocedal, Daniel Robinson, Luis Nunes Vicente M.S. in Industrial & Systems Engineering	Aug. 2018 – May 2022 Aug. 2016 – May 2018
	Shanghai Jiao Tong University B.S.E. in Mechanical Engineering	Aug. 2012 – Aug. 2016
RESEARCH EXPERIENCE	Research Assistant Lehigh University , Bethlehem, PA <i>Department of Industrial & Systems Engineering</i>	Aug. 2018 – May 2022
	Research Intern Facebook, Inc. , New York, NY <i>Facebook AI Research</i>	May 2021 – Aug. 2021
	Givens Associate Argonne National Laboratory , Lemont, IL <i>Mathematics and Computer Science Division</i>	May 2020 – Aug. 2020
	Visiting Research Assistant Northwestern University , Evanston, IL <i>Department of Industrial Engineering and Management Sciences</i>	June 2017 – Aug. 2017

**PUBLISHED
JOURNAL
ARTICLES**

- [1] Albert S. Berahas, Miaolan Xie, and **Baoyu Zhou***. A Sequential Quadratic Programming Method with High-Probability Complexity Bounds for Nonlinear Equality-Constrained Stochastic Optimization. *SIAM Journal on Optimization*, 35(1):240-269, 2025.
- [2] Frank E. Curtis, Daniel P. Robinson, and **Baoyu Zhou***. Sequential Quadratic Optimization for Stochastic Optimization with Deterministic Nonlinear Inequality and Equality Constraints. *SIAM Journal on Optimization*, 34(4):3592-3622, 2024.
- [3] Frank E. Curtis, Daniel P. Robinson, and **Baoyu Zhou***. A Stochastic Inexact Sequential Quadratic Optimization Algorithm for Nonlinear Equality-Constrained Optimization. *INFORMS Journal on Optimization*, 6(3-4):173-195, 2024.
- [4] Albert S. Berahas, Jiahao Shi, Zihong Yi, and **Baoyu Zhou***. Accelerating Stochastic Sequential Quadratic Programming for Equality Constrained Optimization using Predictive Variance Reduction. *Computational Optimization and Applications*, 86:79-116, 2023.
- [5] Albert S. Berahas, Frank E. Curtis, and **Baoyu Zhou***. Limited-Memory BFGS with Displacement Aggregation. *Mathematical Programming*, 194(1):121-157, 2022.
- [6] Jeffrey Larson, Matt Menickelly, and **Baoyu Zhou***. Manifold Sampling for Optimizing Nonsmooth Nonconvex Compositions. *SIAM Journal on Optimization*, 31(4):2638-2664, 2021.
- [7] Albert S. Berahas, Frank E. Curtis, Daniel P. Robinson, and **Baoyu Zhou***. Sequential Quadratic Optimization for Nonlinear Equality Constrained Stochastic Optimization. *SIAM Journal on Optimization*, 31(2):1352-1379, 2021.
- [8] Frank E. Curtis, Daniel P. Robinson, and **Baoyu Zhou***. A Self-Correcting Variable-Metric Algorithm Framework for Nonsmooth Optimization. *IMA Journal of Numerical Analysis*, 40(2):1154-1187, 2020.

**PUBLISHED
CONFERENCE
ARTICLES**

- [9] Jason C. N. Liang, Haihao Lu, and **Baoyu Zhou***. Online Ad Procurement in Non-stationary Autobidding Worlds. *Advances in Neural Information Processing Systems*, 36:42552-42575, 2023.

**WORKING
PAPERS**

- [10] Jiachen Liu, Jiaxin Pei, Jintao Huang, Chenglei Si, Ao Qu, Xiangru Tang, Runyu Lu, Lichang Chen, Xiaoyan Bai, Haizhong Zheng, Carl Chen, Zhiyang Chen, Haojie Ye, Yajuan Fu, Zexue He, Zijian Jin, Zhenyu Zhang, Shangquan Sun, Maestro Harmon, John Dianzhuo Wang, Jianqiao Zeng, Jiachen Sun, Mingyuan Wu, **Baoyu Zhou**, Yuchen You, Shijian Lu, Yiming Qiu, Fan Lai, Yuan Yuan, Yao Li, Junyuan Hong, Ruihao Zhu, Beidi Chen, Alex Pentland, Ang Chen, Mosharaf Chowdhury, and Zechen Zhang. The Last Human-Written Paper: Agent-Native Research Artifacts. arXiv 2604.24658, 2026.
Accepted (Spotlight Presentation) by AI4Science Workshop at 2026 International Conference on Machine Learning
Accepted (Poster) by AI4Research Workshop at 2026 International Conference on Machine Learning
Accepted (Oral) by AgentSkills Workshop at 2026 ACM Conference on AI and Agent Systems
- [11] Luwei Bai[†], Yang Zeng, and **Baoyu Zhou***. Generalization of Silver Stepsize Schedule to Stochastic Optimization. arXiv 2511.21917, 2025.
- [12] Haoming Shen, Yang Zeng, and **Baoyu Zhou***. Sequential Quadratic Optimization for Solving Expectation Equality Constrained Stochastic Optimization Problems. arXiv 2503.09490, 2025.
- [13] Albert S. Berahas, Jiahao Shi, and **Baoyu Zhou***. Optimistic Noise-Aware Sequential Quadratic Programming for Equality Constrained Optimization with Rank-Deficient Jacobians. arXiv 2503.06702, 2025.

All publications with “” sign on my name indicates the author list is arranged in alphabetical order.

[†]Students under my supervision are underlined.

- [14] John R. Birge, Haihao Lu, and **Baoyu Zhou***. On the Convergence of L-shaped Algorithms for Two-Stage Stochastic Programming. *arXiv 2309.01244*, 2023.
- [15] Aaron Defazio, **Baoyu Zhou**, and Lin Xiao. Grad-GradaGrad? A Non-Monotone Adaptive Stochastic Gradient Method. *arXiv 2206.06900*, 2022.
- [16] Albert S. Berahas, Raghu Bollapragada, and **Baoyu Zhou***. An Adaptive Sampling Sequential Quadratic Programming Method for Equality Constrained Stochastic Optimization. *arXiv 2206.00712*, 2022.

DISSERTATIONS

- [17] **Baoyu Zhou**. Methods for Large Scale Nonlinear Optimization and Equality Constrained Stochastic Optimization. Ph.D. Thesis, Department of Industrial & Systems Engineering, Lehigh University, Bethlehem, PA, USA, 2022.
- [18] **Baoyu Zhou**. Quadratic Optimization for Nonsmooth Optimization Algorithms: Thoery and Numerical Experiments. Master Thesis, Department of Industrial & Systems Engineering, Lehigh University, Bethlehem, PA, USA, 2018.

PRESENTATIONS

- [1] Baoyu Zhou. Generalization of Silver Stepsize Schedule to Stochastic Optimization. *INFORMS Optimization Society Conference, Atlanta, GA, USA*, March 2026.
- [2] Baoyu Zhou. Adaptive Stochastic Algorithms for Nonconvex Constrained Optimization. *International Conference on Continuous Optimization, Los Angeles, CA, USA*, July 2025.
- [3] Baoyu Zhou. Inexact SQP Method for Nonlinear Constrained Stochastic Optimization. *INFORMS Annual Meeting, Seattle, WA, USA*, October 2024.
- [4] Baoyu Zhou. An Efficient Algorithm for Large-Scale Stochastic Optimization with Constraints. *INFORMS Annual Meeting, Seattle, WA, USA*, October 2024.
- [5] Baoyu Zhou. Adaptive Stochastic Algorithms for Nonconvex Constrained Optimization. *School of Mathematical and Statistical Sciences, Arizona State University, Tempe, AZ, USA*, September 2024.
- [6] Baoyu Zhou. New Understandings on L-shaped Methods for Two-Stage Stochastic Programming. *INFORMS Optimization Society Conference, Houston, TX, USA*, March 2024.
- [7] Baoyu Zhou. Next Generation Algorithms for Constrained Stochastic Optimization. *School of Computing and Augmented Intelligence, Arizona State University, Tempe, AZ, USA*, January 2024.
- [8] Baoyu Zhou. Next Generation Algorithms for Constrained Stochastic Optimization. *Department of Economics and Business, Colorado School of Mines, Golden, CO, USA*, January 2024.
- [9] Baoyu Zhou. Next Generation Algorithms for Constrained Stochastic Optimization. *Operations Research and Industrial Engineering Program, The University of Texas at Austin, Austin, TX, USA*, January 2024.
- [10] Baoyu Zhou. Next Generation Algorithms for Constrained Stochastic Optimization. *Department of Mathematics, Florida State University, Tallahassee, FL, USA*, January 2024.
- [11] Baoyu Zhou. New Understandings on L-shaped Methods for Two-Stage Stochastic Programming. *INFORMS Annual Meeting, Phoenix, AZ, USA*, October 2023.
- [12] Baoyu Zhou. New Understandings on L-shaped Methods for Two-Stage Stochastic Programming. *Cornell Young Researcher Workshop, Ithaca, NY, USA*, October 2023. (poster)
- [13] Baoyu Zhou. Methods for Constrained Stochastic Optimization: Theory and Practice. *Argonne National Laboratory, Lemont, IL, USA*, September 2023.
- [14] Baoyu Zhou. Inexact Bundle Methods for Two-Stage Stochastic Programming. *MOPTA Conference, Bethlehem, PA, USA*, August 2023.

- [15] Baoyu Zhou. Inexact Bundle Methods for Two-Stage Stochastic Programming. *School of Mathematics, The University of Edinburgh, Edinburgh, Scotland*, July 2023.
- [16] Baoyu Zhou. Inexact Bundle Methods for Two-Stage Stochastic Programming. *SIAM Conference on Optimization, Seattle, WA, USA*, June 2023.
- [17] Baoyu Zhou. SQP Methods for Deterministically Constrained Stochastic Optimization. *Department of Systems and Industrial Engineering, The University of Arizona, Tucson, AZ, USA*, February 2023.
- [18] Baoyu Zhou. SQP Methods for Inequality Constrained Stochastic Optimization. *INFORMS Annual Meeting, Indianapolis, IN, USA*, October 2022.
- [19] Baoyu Zhou. SQP Methods for Deterministically Constrained Stochastic Optimization. *FutureBAProf Workshop, The University of Iowa, Iowa City, IA, USA*, August 2022.
- [20] Baoyu Zhou. SQP Methods for Inequality Constrained Stochastic Optimization. *International Conference on Continuous Optimization, Bethlehem, PA, USA*, July 2022.
- [21] Baoyu Zhou. Fast, Efficient and Practical Algorithms for Nonlinear Optimization. *Department of Industrial, Manufacturing and Systems Engineering, Texas Tech University, Lubbock, TX, USA*, December 2021.
- [22] Baoyu Zhou. An Inexact Sequential Quadratic Method For Nonlinear Equality Constrained Stochastic Optimization. *INFORMS Annual Meeting, Anaheim, CA, USA*, October 2021.
- [23] Baoyu Zhou. Manifold Sampling for Optimizing Nonsmooth Nonconvex Compositions. *MOPTA Conference, Bethlehem, PA, USA*, August 2021.
- [24] Baoyu Zhou. SQP for Nonlinear Equality Constrained Stochastic Optimization. *MOPTA Conference, Bethlehem, PA, USA*, August 2021.
- [25] Baoyu Zhou. SQP Methods for Equality Constrained Stochastic Optimization. *SIAM Conference on Optimization, Spokane, WA, USA*, July 2021.
- [26] Baoyu Zhou. Manifold Sampling for Optimizing Nonsmooth Nonconvex Compositions. *SIAM Conference on Computational Science and Engineering, Fort Worth, TX, USA*, March 2021.
- [27] Baoyu Zhou. Manifold Sampling for Optimizing Nonconvex Piecewise-Smooth Compositions. *INFORMS Annual Meeting, National Harbor, MD, USA*, November 2020.
- [28] Baoyu Zhou. Limited-Memory BFGS with Displacement Aggregation. *INFORMS Annual Meeting, Seattle, WA, USA*, October 2019.
- [29] Baoyu Zhou. Limited-Memory BFGS with Displacement Aggregation. *MOPTA Conference, Bethlehem, PA, USA*, August 2019.

RESEARCH FUNDING

Grants Awarded

- [1] *Collaborative Research: Noise-Aware Optimization Algorithms for Hybrid Quantum Computing*
 Sponsor: National Science Foundation
 Role: Lead-PI
 Team: X. Yang (Lehigh University, PI)
 Total Budget: \$299,998; My Share: 67% (\$199,998)
 Effective: July 2026 – June 2029
- [2] *GPU-Accelerated Large-Scale Optimization for Intensity Modulated Proton Therapy*
 Sponsor: Mayo Clinic
 Role: Lead-PI
 Team: W. Liu (Mayo Clinic Arizona, PI)
 Total Budget: \$22,264; My Share: 100% (\$22,264)
 Effective: June 2026 – August 2026

- [3] *Development of Multi-Scale Traffic Flow Simulation Model for Urban Mobility Applications*
 Sponsor: Argonne National Laboratory, Department of Energy
 Role: Co-PI
 Team: X. Zhou (Arizona State University, PI), H. Wei (Arizona State University, Co-PI)
 Total Budget: \$75,002; My Share: 20% (\$15,000)
 Effective: February 2026 – January 2027

TEACHING EXPERIENCE

Instructor

Arizona State University, Tempe, AZ

IEE 620: Optimization I

Teaching Evaluation: 4.91/5.00 (School Mean [Standard Deviation]: 4.56 [0.31]) Fall 2025

IEE 470: Stochastic Operations Research

Teaching Evaluation: 4.37/5.00 (School Mean [Standard Deviation]: 4.32 [0.45]) Spring 2026

Teaching Evaluation: 4.41/5.00 (School Mean [Standard Deviation]: 4.13 [0.62]) Spring 2025

Guest Lecturer

University of Michigan, Ann Arbor, MI

IOE 691: Advanced Computational Optimization Winter 2024

Bootcamp Organizer & Instructor

Lehigh University, Bethlehem, PA

Aug. 2021

Teaching Assistant Mentor

Shanghai Jiao Tong University, Shanghai, China

Sept. 2014 – Aug. 2016

Undergraduate Teaching Assistant

Shanghai Jiao Tong University, Shanghai, China

VV156: Honors Calculus II

Fall 2013, Spring 2014, Fall 2015

VM240: Intro to Dynamics and Vibrations

Summer 2015

HONORS & AWARDS

Alliance Fellow, Mayo Clinic-ASU Faculty Summer Residency Program

2026

Elizabeth V. Stout Dissertation Award, Lehigh University

2022

SIAM Student Travel Award

2021

Van Hoesen Family Best Publication Award, Lehigh ISE Department

2021

P.C. Rossin Doctoral Fellow, Lehigh University

2021

Ph.D. Student of the Year, Lehigh ISE Department

2021

Lehigh University Fellowship

2018 – 2019

Outstanding Freshman Scholarship, Shanghai Jiao Tong University

2012 – 2016

Outstanding Academic Scholarship, Shanghai Jiao Tong University

2015

MENTORSHIP EXPERIENCE

Ph.D. Student Advisees

• Yunfan Zeng (Arizona State University)

2025 – present

• Luwei Bai (Arizona State University)

2025 – present

• Yang Zeng (Arizona State University)

2024 – present

Doctoral Committee Membership (outside my research group)

• Henan Zhu (Arizona State University)

2026 – present

Advisor: Xuesong Zhou

- Kamrun Nahar Keya (Arizona State University) 2026 – present

Advisor: Johannes Brust

- Lu Liu (Arizona State University) 2025 – present

Advisor: Feng Ju

- Qi Zhang (Arizona State University) 2025 – 2026

Topic: *Distributionally Robust Optimization for Machine Learning*

Advisor: Shaofeng Zou

Master Student Advisees

- Ashish Goud Balagula (Arizona State University) 2025 – present

ACADEMIC SERVICE

Professional Affiliations

- Society for Industrial and Applied Mathematics (SIAM) 2021 – present
- Institute for Operations Research and Management Sciences (INFORMS) 2018 – present

Professional Community Service

- Judge, INFORMS Railway Applications Section Problem Solving Competition 2026
- Panelist, Lehigh ISE Future Optimization Graduate Student Workshop 2026
- Reviewer, INFORMS DEI Best Student Paper Award 2024
- Vice President, Lehigh University INFORMS Student Chapter 2020 – 2021

Organized Conference Sessions

- **ICCOPT Conference**

- *Methods for Large-Scale Nonlinear Optimization I-V*, Los Angeles, CA, USA, July 2025.
- *Advances in Nonlinear Optimization*, Bethlehem, PA, USA, July 2022.
- *Large-Scale, Nonlinear, and Stochastic Optimization IV*, Bethlehem, PA, USA, July 2022.

- **INFORMS Annual Meeting**

- *Advances in Nonlinear, Stochastic, and Constrained Optimization*, Seattle, WA, USA, October 2024.
- *Recent Advances in Nonlinear and Stochastic Optimization*, Indianapolis, IN, USA, October 2022.
- *Nonlinear and Stochastic Optimization*, Indianapolis, IN, USA, October 2022.
- *Advances in Nonlinear and Stochastic Optimization (I–II)*, Anaheim, CA, USA, October 2021.
- *Optimization in Quantum Computing and vice versa I*, Anaheim, CA, USA, October 2021.

- **INFORMS Optimization Society Conference**

- *Methods for Large-Scale Nonlinear and Stochastic Optimization (I–III)*, Houston, TX, USA, March 2024.
- *Recent Advances in Nonlinear Stochastic Optimization and Related Applications*, Atlanta, GA, USA, March 2026.

- **International Symposium on Mathematical Programming**

- *Advances in Large-Scale Nonlinear Optimization (I–IV)*, Montréal, Canada, July 2024.

- **MOPTA Conference**

- *Advances in Nonlinear and Stochastic Optimization*, Bethlehem, PA, USA, August 2026.

- *Nonlinear and Stochastic Optimization Algorithms*, Bethlehem, PA, USA, August 2021.
- *Algorithms for Derivative-Free Optimization*, Bethlehem, PA, USA, August 2021.
- **SIAM Conference on Optimization**
- *Recent Advances in Large-Scale Nonlinear Optimization I-II (Sessions I-IV)*, Seattle, WA, USA, May 2023.

Journals Reviewed for (# of articles reviewed)

- Computational Optimization and Applications (9)
- EURO Journal on Computational Optimization (1)
- IEEE Control Systems Letters (1)
- IEEE Transactions on Automatic Control (1)
- IEEE Transactions on Automation Science and Engineering (1)
- IEEE Transactions on Network Science and Engineering (1)
- IEEE Transactions on Neural Networks and Learning Systems (4)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (1)
- IMA Journal of Numerical Analysis (1)
- INFORMS Journal on Computing (1)
- INFORMS Journal on Data Science (1)
- International Journal of Radiation Oncology, Biology, Physics (2)
- Journal of Global Optimization (1)
- Journal of Machine Learning Research (1)
- Journal of Optimization Theory and Applications (1)
- Journal of Scientific Computing (3)
- Machine Learning (1)
- Management Science (2)
- Mathematical Methods of Operations Research (1)
- Mathematical Programming (7)
- Mathematical Programming Computation (1)
- Mathematics and Computers in Simulation (1)
- Mathematics of Computation (1)
- Mathematics of Operations Research (1)
- Numerical Algorithms (1)
- Operations Research (1)
- Optimization Methods and Software (4)
- SIAM Journal on Mathematics of Data Science (2)
- SIAM Journal on Matrix Analysis and Applications (1)
- SIAM Journal on Optimization (6)
- Signal Processing (1)

Conferences Reviewed for (# of articles reviewed)

Serving as Area Chair

- Constrained Optimization for Machine Learning (COML) Workshop, NeurIPS 2025 (6)

Serving as Program Committee (Reviewer)

- INFORMS Workshop on Data Mining and Decision Analytics 2026
- Refereed Proceedings, INFORMS Optimization Society Conference 2026 (1)
- When Machine Learning meets Dynamical Systems: Theory and Applications (MLmDS) Workshop, AAAI 2023 (1)
- Optimization for Machine Learning (OPT) Workshop, NeurIPS 2022 (3)
- New Frontiers in Federated Learning (NFFL) Workshop, NeurIPS 2021 (3)
- Optimization for Machine Learning (OPT) Workshop, NeurIPS 2021 (3)
- Optimization for Machine Learning (OPT) Workshop, NeurIPS 2020 (4)

COURSES

Convex Analysis; Planning & Scheduling in Manufacturing & Services; Optimization Models & Applications; Nonlinear Optimization; Random Processes & Applications; Intro to Math Optimization; Real Analysis I; Discrete Optimization; Computational Methods in Optimization; Dynamic Programming; Applied Operations Research; Optimization Methods in Machine Learning; Mining of Large Datasets; Quantum Computing for Optimization